

# Arunan Jithendra Kumar

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## Summary

Computer Science graduate with production software engineering experience at Amazon, building and deploying backend systems, APIs, data pipelines, and ML-powered services in Python. Experienced in production testing, asynchronous workflows, cloud deployment, performance optimization, and end-to-end system development. Improved production accuracy by **22.2 percentage points**, reduced median inference latency by **~50%**, and reduced per-request token consumption by **98.7%**. Foundation in data structures, algorithms, operating systems, databases, and computer networks.

## Technical Skills

**Languages:** Python, Java, C, SQL, Bash, React, Next.js

**Backend and Software Engineering:** Node.js, FastAPI, Flask, REST APIs, Async Programming, Object Oriented Programming, Automated Testing, Git, CI/CD, Linux, Maven, Jenkins, Junit

**Cloud & Deployment:** AWS Bedrock, SageMaker, S3, Docker

**Data & Distributed:** PySpark, NumPy, Pandas, Data Processing Pipelines

**AI/ML:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LoRA, SFT, RAG, Model Distillation

**Additional:** React, MLflow, LlamaIndex, vLLM, TensorRT

## Education

**Shiv Nadar University, Chennai**

Sept 2022 – July 2026

B.Tech in Computer Science (Specialization in Internet of Things)

**CGPA:** 8.3/10

**Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks

## Experience

**Applied Scientist Intern – Amazon (Devices Org, Alexa Connections)**

Mar 2026 – Jul 2026

*Production LLM and Small Language Model Systems for Alexa+ AnnouncementExpert*

- Designed and deployed an end-to-end **Python** production workflow for Alexa+ AnnouncementExpert, replacing prompt-driven expert orchestration with a specialized model-serving pipeline using **LoRA**-based SFT for **Llama 3.2 1B**.
- Improved production accuracy from **74.8% to 97.0%** (+22.2 percentage points) while reducing median inference latency by **~50%** on the same hardware.
- Reduced the request prompt footprint from **~4,877 instruction tokens** to **~66 input tokens**, achieving a **98.7% reduction** in token consumption per request.
- Built automated **evaluation and testing pipelines** over production utterance distributions to validate functional correctness, quality, latency, and safety before deployment.
- Developed the model workflow to directly handle **JSON/API generation, content extraction, ask-for-content, and cancellation flows**, eliminating DialogManager orchestration overhead.
- Used internal **Git and pipeline tooling** for development, testing, integration, and deployment; packaged and deployed custom checkpoints to **AWS Bedrock via Custom Model Import**.
- Authored production-readiness recommendations covering monitoring, fallback strategies, retraining cadence, and deployment considerations for continued operation.

**Project Staff – Speech and Music Technology Lab, Dept. of CSE, IIT Madras**

May 2025 – Feb 2026

*Ministry of Electronics & IT (Govt. of India)-funded Project*

- Designed and implemented a hybrid **SimCLR**-based self-supervised learning pipeline using CFCC representations and DTW-based embeddings for microtonal acoustic signals.
- Delivered a **91% relative improvement** over the lab's internal baseline on downstream evaluation; first-author manuscript under review at **ISMIR 2026**.

*Neural Networks without Backpropagation for ASR*

- Implemented and evaluated **Random Feedback Alignment (RFA)** and **Direct Feedback Alignment (DFA)** across MNIST, CIFAR-10, and TIMIT.
- Demonstrated on-par performance with backpropagation across shallow and deep architectures; co-authored manuscript under review at **Interspeech 2026** (equal contribution).

**Applied AI & Cloud Data Engineering Intern – NIT Trichy**

Apr 2025 – May 2025

- Built and deployed a **Dockerized Flask** time-series forecasting service for automated cloud resource autoscaling, integrating anomaly detection and data-drift monitoring for streaming workloads.
- Automated data ingestion and **JSON/schema validation** workflows, reducing manual processing effort by **60%**.
- Worked on the distributed-system architecture and system design of the time-series forecasting service; developed unit and integration tests to validate service behavior and support debugging.

## Selected Projects

**Dynamic RAG Schema Designer – Autonomous Retrieval Optimization System**

- Built a backend engine that automatically generates, benchmarks, and selects retrieval configurations across chunking, embeddings, and reranking strategies, eliminating manual RAG configuration.
- Developed a **FastAPI + React** experimentation platform with real-time analytics and support for multiple vector backends.

**ADCEA – Autonomous Data Cleaning & Feature Engineering Agent**

- Built an end-to-end data processing application for dataset ingestion, profiling, cleaning, feature engineering, and baseline AutoML training.
- Implemented **asynchronous training pipelines**, Argon2-based authentication, and a lightweight SPA for live insights and model management.

## Research Publications

- A. Jithendra Kumar** et al., “Detecting Neural Network Failures through Spectral Analysis of Internal Activations,” submitted to **ACML 2026**. *First author*.
- A. Jithendra Kumar** & co-author, “Deep Neural Networks for Automatic Speech Recognition Without Backpropagation,” submitted to **Interspeech 2026**. *Equal contribution*.

## Certifications

Responsible & Safe AI Systems (NPTEL) | Embedded Systems (IIT Madras) | ML & DS Mathematics (NIT Trichy)  
| Google Cloud Data Analytics | AWS Cloud Practitioner, GenAI, Agentic AI

